Bandit Problems with Causal Background Knowledge

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Topics

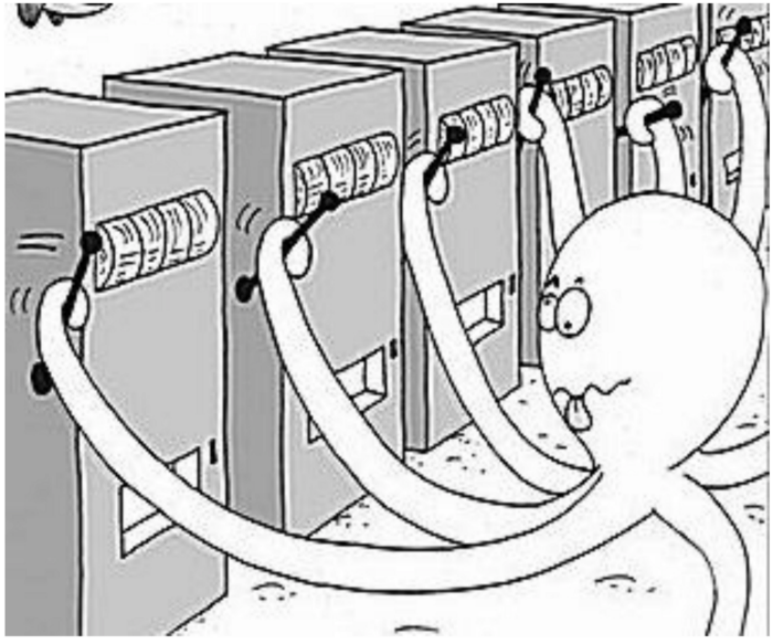
- Lattimore et al., *Causal Bandits: Learning Good Interventions via Causal Inference*, NIPS 2016.
- Lu et al., *Regret Analysis of Bandit Problems with Causal Background Knowledge*, UAI 2020.

1. Bandits
2. Applications
3. Causal Bandits
4. Causal Bandit Algorithms
5. Experiments
6. Challenges
7. C-UCB Regret Bound Proof

Section 1

Bandits

Multi-Armed Bandits



Bandits vs. RL (MDP)?

Section 2

Applications

Applications

- **Email Campaign Problem**: Convert a potential buyer to actual buyer. Maximize user interaction with our site.
- Parameters:
  - Length of the subject
  - Time of the day the email was sent
  - Template
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- Parameters:
  - Length of the subject
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  - Template
  - With links or without links ...
- Exploration vs. Exploitation!
- The number of possible actions in huge. Thus we need to consider the relations between these actions and the cannot be treated separately.
- **Causal Model**?

Section 3

Causal Bandits

Causal Bandits: Problem Setup

- We are given a causal model $\mathcal{C}$ and graph $\mathcal{G}$.
- All variables $\mathcal{X} = \{X_1, \ldots, X_N\}$ are observable.
- There is a reward variable Y.
- Each variable has a discrete alphabet of size k.
- Denote PA_Y at round t as $\mathrm{Z}_{(t)} = (z_1(t), x_2(t), \ldots, z_n(t))^\top$.
- There are k^n assignments for Z, denoted as $\mathrm{Z}^1, \ldots, \mathrm{Z}^{k^n}$.

Causal Bandits: Problem Setup

- The learner is given the graph, $\mathcal{G}$, a set of possible interventions $\mathcal{A}$, and all inteventional distributions $P(\mathrm{PA}_Y|a)$ for $a \in \mathcal{A}$.
- It is assumed that $\mathcal{A}$ only consists of single node or zero-node interventions.

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- It is assumed that $\mathcal{A}$ only consists of single node or zero-node interventions.
- Expected reward of each action a and the optimal action a^*:

$$\mu_a = \mathbb{E}[Y|a] \in [0,1]$$
$$a^* = \arg\max_{a \in \mathcal{A}} \mu_a$$

Causal Bandits: The Sequential Game

- In round t, the learner chooses the action a_t based on the observation of $Y_1, \ldots, Y_t$ and $Z_1, \ldots, Z_{t-1}$ in previous rounds.

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- **Goal:** Maximize cumulative regret:

$$\mathbb{E}[R_T] = T\mu^* - \sum_{t=1}^{T} \mathbb{E}[\mu_{a_t}]$$

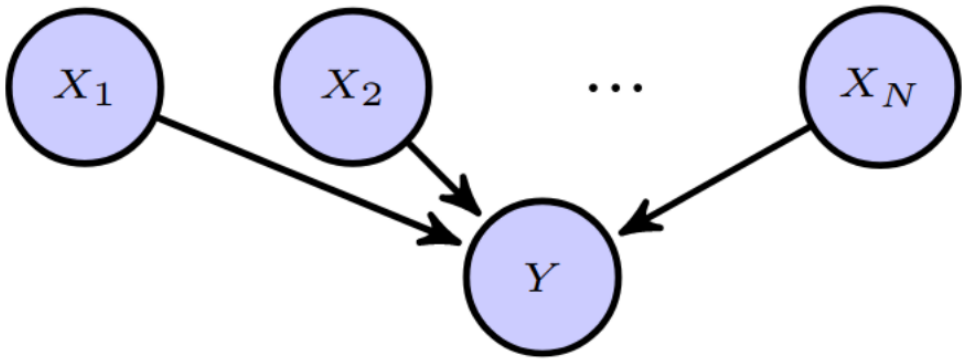
Section 4

Causal Bandit Algorithms

Simple Algorithms for Some Intuition

In most bandit problems, the **Explore-Then-Commit** algorithm is analyzed first:

- Perform each possible intervention L times. Estimate means $\hat{\mu}_a$.
- Sample from $\hat{a}^* = \arg\max_{a \in \mathcal{A}} \hat{\mu}_a$ for $T - L$ rounds.



Causal Upper Confidence Bound

For $t = 1, \ldots, T$:

1. $\forall j = 1, \ldots, k^n$:

$$\mathrm{UCB}_{\mathrm{Z}_j}(t-1) = \hat{\mu}_{\mathrm{Z}^j}(t-1) + \sqrt{\frac{2\log(1/\delta)}{1 \vee T_{\mathrm{Z}^j}(t-1)}}$$

2. $a_t = \arg\max_{a \in \mathcal{A}} \sum_{j=1}^{k^n} \mathrm{UCB}_{\mathrm{Z}^j}(t-1) P(\mathrm{Pa}_Y = \mathrm{Z}^j | a)$
3. Pull arm a_t and observe reward Y_t and its parent nodes' values $\mathrm{Z}_{(t)}$.
4. $\forall j = 1, \ldots, k^n$, update

$$T_{\mathrm{Z}^j}(t) = \sum_{s=1}^{t} 1_{\{\mathrm{Z}_s = \mathrm{Z}^j\}}$$

$$\hat{\mu}_{\mathrm{Z}^j}(t) = \frac{1}{T_{\mathrm{Z}^j}(t)} \sum_{s=1}^{t} Y_s 1_{\{\mathrm{Z}_s = \mathrm{Z}^j\}}$$

C-UCB Regret Bound

Theorem

Let $Y|_{PA_Y=Z^{(j)}} = \mathbb{E}[Y|PA_Y = Z^{(j)}] + \epsilon$, *where* ϵ *is a mean zero 1-subgaussian random variable. If* $\delta = 1/T^2$, *the regret of C-UCB is bounded as follows:*

$$\mathbb{E}[R_T] = \tilde{O}\left(\sqrt{k^n T}\right) \tag{1}$$

We will prove this in the end (if we have time).

Causal and Non-Causal Bandits

- **Discarding causal information**:
  There are $(k+1)^N$ interventions. From well known Bandit theory, $R_T = \Omega\left(\sqrt{(k+1)^N T}\right)$. UCB and Thompson sampling achieve this.
- **This Paper**: $R_T = O\left(\sqrt{(k+1)^n T}\right)$
  - Algorithm: a variant of UCB and Thompson Sampling
  - Assumes that we have access to $P(PA_Y|a)$ for all $a \in \mathcal{A}$.
  - Lower bounds are not provided.

Thompson Sampling

For $t = 1, \ldots, T$

- Sample $\hat{\theta}_j(t) \sim N(\hat{\mu}_{Z_j}, \frac{1}{k_{Z_j}+1})$,for $j = 1, \ldots, k^n$.
- $\forall a \in \mathcal{A}: \quad \hat{\mu}_a = \sum_{j=1}^{k^n} \hat{\theta}_j(t) P(\mathrm{Pa}_Y = Z_j | a)$
- $a_t = \arg\max_a \hat{\mu}_a$
- Pull arm a_t and observe the parent nodes values of Y denoted by $Z_{(t)}$ and reward Y_t.
- Update $k_{Z_t} := k_{Z_t} + 1$
- Update $\hat{\mu}_{Z_t} := \frac{\hat{\mu}_{Z_t} k_{Z_t} + Y_t}{k_{Z_t}+1}$

Causal Linear Bandits

- Assume that $Y|_{\mathsf{Z}} = f(\mathsf{Z})^\top \theta + \epsilon$.
- **Idea**: Regression + Confidence bound for regression + UCB algorithm.

Causal Linear Bandits

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Theorem

If $||\theta||_2 \leq 1$, $||f(Z)||_2 \leq 1$, $\theta \in \mathbb{R}^d$:

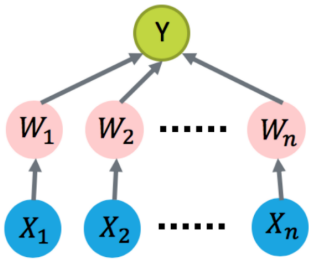
$$R_T = \tilde{O}(d\sqrt{T})$$

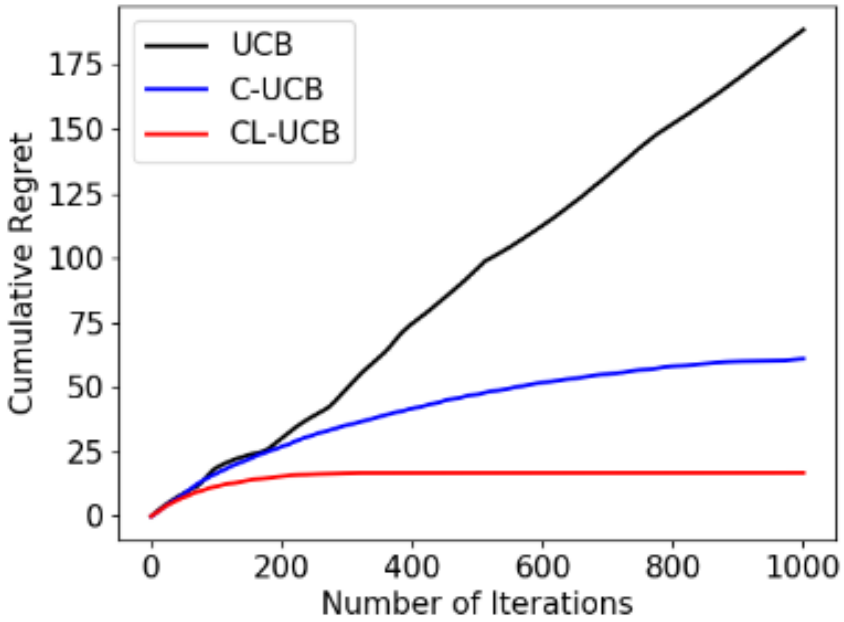
Section 5

Experiments

Pure Simulation

- Fixed conditional probabilities.
- Linear relation between W_is and Y.

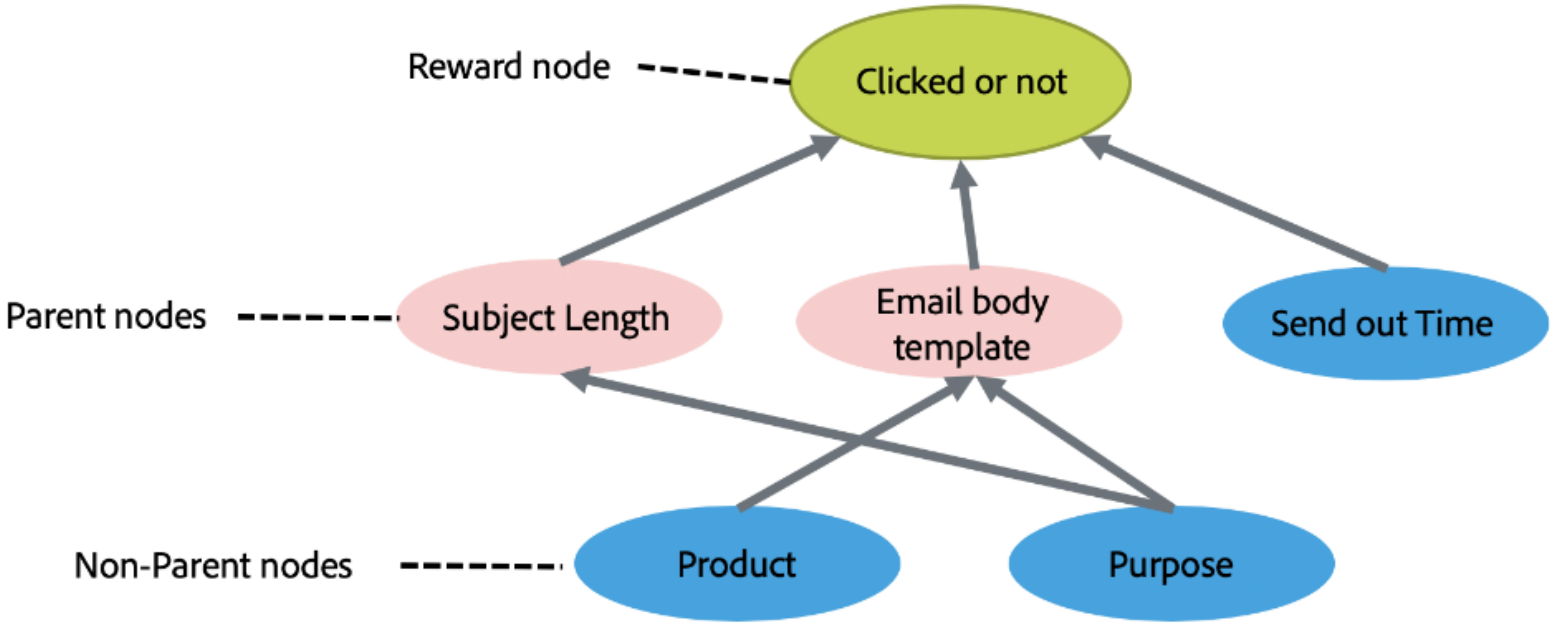




i	1	2	3
$P(X_1 = i)$	0.3	0.4	0.3
$P(X_2 = i)$	0.3	0.3	0.4
$P(X_3 = i)$	0.5	0.3	0.2
$P(X_4 = i)$	0.25	0.25	0.5
$P(W_1 = 1 \mid X_1 = i)$	0.2	0.5	0.8
$P(W_2 = 1 \mid X_2 = i)$	0.3	0.2	0.8
$P(W_3 = 1 \mid X_3 = i)$	0.4	0.6	0.5
$P(W_4 = 1 \mid X_4 = i)$	0.3	0.5	0.6

Real World Simulation!

EMAIL CAMPAIGN DATA



Section 6

Challenges

Challenges

- A much more difficult generalisation would be to consider causal bandit problems where the causal graph is completely unknown or known to be a member of class of models.
- Work on the problem of selecting experiments to discover the correct causal graph from within a Markov equivalence class should be incorporated into a causal bandit algorithm.
- Estimating Interventional Distributions.

Section 7

C-UCB Regret Bound Proof

C-UCB Regret Bound

Now, we will prove the C-UCB regret bound theorem.

- Let E be the following event:

$$\forall t \in [T], j \in [k^n], \quad \left|\hat{\mu}_{Z^j}(t-1) - \mathbb{E}\left[Y | Pa_Y = Z^j\right]\right| \leq \sqrt{\frac{2\log(1/\delta)}{1 \vee T_{Z^j}(t-1)}}.$$

- Using union bound and subgaussanity of ϵ: $\mathbb{P}[E^c] \leq 2\delta T k^n$.

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- Using union bound and subgaussanity of ϵ: $\mathbb{P}[E^c] \leq 2\delta T k^n$.
- Given that event E occurs, we can write:

$$\mathrm{UCB}_{Z_j}(t-1) = \hat{\mu}_{Z^j}(t-1) + \sqrt{\frac{2\log(1/\delta)}{1 \vee T_{Z^j}(t-1)}} \geq \mathbb{E}\left[Y | Pa_Y = Z^j\right]$$

C-UCB Regret Bound

Now we can start bounding the regret:

$$
\begin{aligned}
R_T &= \sum_{t=1}^{T} (\mu_{a^*} - \mu_{a_t}) \\
&= \sum_{t=1}^{T} \left(\mu_{a^*} - \mathrm{UCB}_{a_t}(t-1) + \mathrm{UCB}_{a_t}(t-1) - \mu_{a_t}\right).
\end{aligned}
$$

C-UCB Regret Bound

Hence, given that event E occurs:

$$
\begin{aligned}
\mu_{a^*} - &\mathrm{UCB}_{a_t}(t-1) \\
&= \sum_{j=1}^{k^n} \mathbb{E}\left[Y | Pa_Y = Z^j\right] P(Pa_Y = Z^j | a^*) - \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(Pa_Y = Z_j | a_t) \\
&\leq \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(Pa_Y = Z^j | a^*) - \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(Pa_Y = Z^j | a_t) \leq 0.
\end{aligned}
$$

C-UCB Regret Bound

Hence, given that event E occurs:

$$\begin{aligned}
&\mu_{a^*} - \mathrm{UCB}_{a_t}(t-1) \\
&\quad = \sum_{j=1}^{k^n} \mathbb{E}\left[Y | Pa_Y = Z^j\right] P(Pa_Y = Z^j | a^*) - \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(Pa_Y = Z_j | a_t) \\
&\quad \leq \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(Pa_Y = Z^j | a^*) - \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(Pa_Y = Z^j | a_t) \leq 0.
\end{aligned}$$

The last equality follows from the way we choose a_t in C-UCB:

$$a_t = \arg\max_{a \in \mathcal{A}} \sum_{j=1}^{k^n} \mathrm{UCB}_{Z^j}(t-1) P(\mathrm{Pa}_Y = Z^j | a)$$

C-UCB Regret Bound

Under E, we have:

$$
\begin{aligned}
R_T &\leq \sum_{t=1}^{T} \left(\mathrm{UCB}_{a_t}(t-1) - \mu_{a_t}\right) \\
&= \sum_{t=1}^{T} \sum_{j=1}^{k^n} \left(\mathrm{UCB}_{Z_j}(t-1) - \mathbb{E}\left[Y | Pa_Y = Z^j\right]\right) P(Pa_Y = Z^j | a_t) \\
&\leq \sum_{t=1}^{T} \sum_{j=1}^{k^n} \sqrt{\frac{8\log(1/\delta)}{1 \vee T_{Z^j}(t-1)}} P(Pa_Y = Z^j | a_t) \\
&\leq \sum_{t=1}^{T} \sum_{j=1}^{k^n} \sqrt{\frac{8\log(1/\delta)}{1 \vee T_{Z^j}(t-1)}} \left(\underbrace{P(Pa_Y = Z^j | a_t) - 1_{\{Z_{(t)} = Z^j\}}} + \underbrace{1_{\{Z_{(t)} = Z^j\}}} \right).
\end{aligned}
$$

C-UCB Regret Bound

- The first part:

$$|X_t - X_{t-1}|^2 = \left| \sum_{j=1}^{k^n} \sqrt{\frac{8\log(1/\delta)}{1 \vee T_{Z^j}(t-1)}} \left(P(Pa_Y = Z^j | a_t) - 1_{\{Z_{(t)} = Z^j\}} \right) \right|^2$$
$$\leq 32 \log(1/\delta).$$

Hence it has the bounded difference property.

- The second part:

$$\sum_{t=1}^{T} \sum_{j=1}^{k^n} \sqrt{\frac{8\log(1/\delta)}{1 \vee T_{Z^j}(t-1)}} 1_{\{Z_{(t)} = Z^j\}} \leq \sum_{j=1}^{k^n} \int_0^{T_{Z^j}(T)} \sqrt{\frac{8\log(1/\delta)}{s}} ds$$
$$\leq \sum_{j=1}^{k^n} \sqrt{32 T_{Z^j}(T) \log(1/\delta)}$$
$$\leq \sqrt{32 k^n T \log(1/\delta)}.$$

C-UCB Regret Bound

Azuma-Hoeffding:

$$P(|X_T| > \sqrt{k^n T \log(T)} \log(T)) \leq \exp\left(-\frac{k^n \log^3(T)}{32 \log(1/\delta)}\right).$$

Take $\delta = 1/T^2$. With probability

$$1 - P(E^c) - \exp\left(-\frac{k^n \log^2(T)}{64}\right) = 1 - 2k^n/T - \exp\left(-\frac{k^n \log^2(T)}{64}\right),$$

the regret can be bounded by

$$R_T \leq 16\sqrt{k^n T \log(T)} \log(T).$$

Thus the expected regret can be bounded by:

$$\mathbb{E}[R_T] \leq P(E^c) \times 2T + \exp\left(-\frac{k^n \log^2(T)}{64}\right) \times 2T + \sqrt{64 k^n T \log(T)} \log(T$$

$$\leq C\sqrt{k^n T \log(T)} \log(T).$$